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Serverless 3D Reconstruction and Spatial Anchoring for Cloud-Native Infrastructure Inspection

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Abstract

While infrastructure asset management increasingly relies on high-resolution drone imagery, existing workflows suffer from fragmented information management and dependence on costly local processing infrastructure. This paper addresses these limitations by using a cloud-native spatial intelligence hub that converts raw inspection imagery into an interactive and queryable three-dimensional information layer. The system integrates a timeout-resilient orchestration layer for photogrammetry pipelines, a multi-user three-dimensional environment for collaborative review, and a PostGIS-backed spatial database that stores defects as georeferenced anchors. We further introduce a spatial anchoring workflow mapping three-dimensional interactions to world coordinates, retrieving context-relevant images via frustum-based visibility scoring. Evaluated on real inspection datasets, the serverless architecture achieved end-to-end reconstruction in under one hour with sub-25 ms query latency. Results indicate that acquisition geometry, particularly oblique convergent viewpoints, is a stronger predictor of reconstruction complexity than image count. This work establishes a reproducible reference architecture, enabling a transition from file-centric documentation to traceable, spatially indexed evidence management for infrastructure Digital Twins.

Keywords: spatial intelligence; serverless computing; infrastructure inspection; Digital Twins; photogrammetry; web-based 3D visualization; cloud-native architecture; semantic spatial anchoring

1. Introduction

The Architecture, Engineering, and Construction (AEC) sector has made significant progress in digitizing design and delivery phases, largely through Building Information Modeling (BIM). However, the Operation and Maintenance (O&M) phase, while constituting approximately 60–80% of an asset's total lifecycle cost, remains plagued by systemic inefficiencies rooted in data fragmentation [1–3].

Infrastructure asset management relies heavily on inspection workflows that generate vast quantities of heterogeneous data, including high-resolution images, unstructured pdf reports, and tabular defect logs stored in isolated spreadsheets [4,5]. The fundamental problem facing facility management is not a lack of data, but a lack of context. In current practice, a structural engineer might identify a shear crack in a raw drone image (File A), log its width in an Excel spreadsheet (File B), and describe its severity in a formal report (File C). As illustrated in Figure 1, this fragmentation results in "dead data": information

Check for updates

Academic Editor: Humberto Varum

Received: 19 May 2026
Revised: 11 June 2026
Accepted: 16 June 2026
Published: 18 June 2026

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trapped in disconnected silos, severing the semantic link between a defect's visual evidence and its spatial location on the asset [1,6].

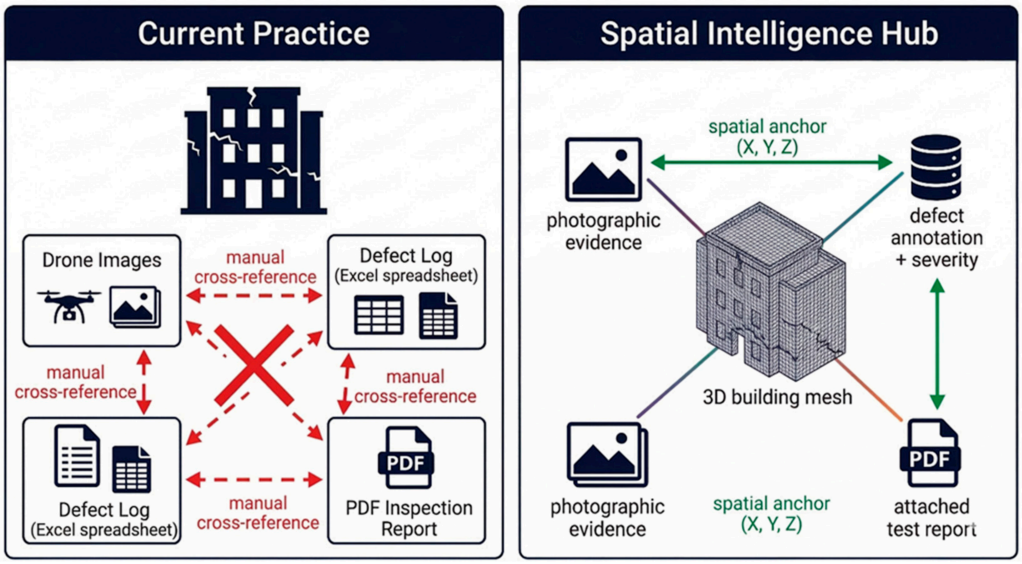

Figure 1. From disconnected inspection data to a spatially anchored intelligence hub.

Empirical studies indicate that while Common Data Environments (CDEs) were intended to solve this via ISO 19650 standards [7–9], current implementations often function merely as "digital filing cabinets", [10] managing files rather than granular information. This disconnect prevents organizations from achieving portfolio-level visibility and inhibits the transition from reactive repairs to predictive maintenance [11,12].

The urgency of this transition is particularly acute in emerging markets, such as Morocco and the broader North African region, where rapid urbanization creates a pressing need to maintain both legacy and modern infrastructure. However, the adoption of high-fidelity Digital Twin technologies in these regions faces an upfront economic barrier: the Capital Expenditure (CapEx) required for local processing hardware.

Traditional photogrammetric workflows and "Scan-to-BIM" processes are computationally punishing. They necessitate high-end local workstations equipped with enterprise-grade GPUs (e.g., NVIDIA RTX series) and expensive perpetual software licenses for Small and Medium Enterprises (SMEs) in emerging markets; this upfront investment is prohibitive [2,13,14]. Furthermore, the workload of an inspection firm is inherently "bursty"; intense data processing is required immediately after a drone flight, followed by periods where expensive hardware sits idle, rendering the fixed-cost workstation model economically inefficient [15–17].

To overcome these hardware limitations, this research posits that emerging markets have a unique opportunity for "Technological Leapfrogging". Just as mobile telephony in Africa bypassed landlines [18,19], construction firms can bypass the "desktop software" phase entirely by adopting cloud-native platforms [20]. By moving from local, file-based workflows to a serverless cloud architecture [21], we can decouple inspection capabilities from local hardware constraints, converting prohibitive CapEx into flexible Operational Expenditure (OpEx).

To enable this transition, we propose the concept of a "Spatial Intelligence Hub". Unlike traditional "Level 1" Digital Twins, which often serve as passive 3D visualization [3,22], this architecture treats the 3D reconstruction as a "spatial organizing substrate" [23–25]. In this paradigm, the distinction matters operationally. When a defect is recorded as an unstructured text description ("north façade, third floor, near expansion joint"), matching it to its physical location requires human interpretation at every subsequent access.

Conversely, when the defect is recorded as a three-dimensional coordinate anchored to a georeferenced mesh surface, the same defect can be retrieved by any spatial query that includes that coordinate regardless of who wrote the original description or when. This approach creates a "Semantic Digital Twin" where geometry and business logic are integrated into a unified system [26]. Consequently, this allows for bidirectional information flow: users can query the model to find defects or query the database to locate them in 3D space [24].

The mathematical foundations of recovering 3D structure from 2D images are well-established through Structure-from-Motion (SfM) [27,28], and Multi-View Stereo (MVS) algorithms [29]. Open-source toolkits such as COLMAP have democratized access to these algorithms, allowing for the generation of high-fidelity sparse and dense point clouds from unordered image collections. While recent advances in Neural Radiance Fields (NeRF) and 3D Gaussian Splatting (3DGS) offer superior visual fidelity for rendering, traditional SfM/MVS remains the gold standard for engineering applications requiring explicit geometric accuracy and metric measurements [30–32]. However, these tools typically operate as monolithic software stacks designed for vertical scaling, necessitating high-end local workstations equipped with enterprise-grade GPUs (e.g., NVIDIA RTX series) and significant RAM. This hardware dependency creates a "Digital Divide", particularly in emerging markets where the high Capital Expenditure (CapEx) for such workstations is prohibitive for Small and Medium Enterprises (SMEs) [2]. Furthermore, because inspection workloads are inherently bursty, requiring intense processing immediately after a flight followed by idle periods, the traditional workstation model represents an economically inefficient allocation of resources [15,16].

While the industry has moved toward Common Data Environments (CDEs) to comply with ISO 19650 standards, empirical studies indicate that current implementations often function merely as digital filing cabinets, managing files rather than granular information. This file-centric approach results in dead data, where defect logs, PDF reports, and 2D images remain semantically disconnected from the geometric reality of the asset.

Although Building Information Modeling (BIM) offers semantic richness, it is fundamentally design-centric, assuming data flows from design to operations. This paradigm is poorly suited for post-construction inspection, where as-built reality diverges from design intent and requires costly Scan-to-BIM manual modeling to reconcile.

Consequently, most infrastructure Digital Twins remain at level 1 maturity serving as passive visual replicas that lack the spatial intelligence to support complex queries or automated maintenance planning [3,33].

Cloud computing offers a potential solution to hardware constraints; however, traditional lift-and-shift strategies migrating virtual machines to the cloud fail to address the cost inefficiencies of idle hardware [15]. Serverless computing (e.g., AWS Lambda, Google Cloud Run) offers a superior pay-for-value model but introduces significant architectural challenges for photogrammetry [16].

Standard serverless containers are stateless and impose strict execution time limits (timeouts) [34], making them historically unsuitable for long-running SfM/MVS pipelines that can take hours to process. While recent work has explored serverless architectures for lightweight geospatial tasks like tile serving, there remains a paucity of research on orchestrating heavy, stateful reconstruction pipelines in a serverless environment [35–37]. An additional gap concerns workload estimation. Existing deployments often approximate processing cost using simple input metadata such as image count or file size, despite photogrammetric literature demonstrating that the computational complexity of Multi-View Stereo is strongly influenced by acquisition geometry, particularly viewing angle diversity and baseline-to-depth ratio, rather than image count or Ground Sampling Distance

alone [38–40]. To the best of our knowledge, the implications of geometry-driven reconstruction complexity for cloud execution planning and serverless timeout management have not been systematically examined.

This paper addresses this gap by proposing an event-driven architecture using asynchronous orchestration to manage long-running photogrammetric workloads without persistent infrastructure (Table 1).

Table 1. Positioning of the proposed platform with respect to common inspection solutions.

Category	Typical Solutions	3D Reconstruction Pipeline	Cloud-Native Execution	Spatial DB (GIS)	3D Defect Anchoring	Multi-User Review	Scalability & Cost Control
Desktop photogrammetry	COLMAP, Metashape, RealityCapture	✓	! (manual/limited)	✗	! (often external)	✗	(CapEx + manual ops)
BIM/CDE platforms	BIM 360/ACC, ISO 19650 workflows	✗	✓	(not spatial-first)	✗	✓	✓
Digital Twin viewers	Cesium-based viewers, web 3D portals	✓ (if imported)	✓	✗	(limited)	✓	✓
Defect management systems	CMMS, ticketing systems	✗	✓	✗	✗	✓	✓
This work	Spatial intelligence hub	✓✓	✓✓ (serverless + orchestration)	✓✓ (PostGIS)	✓✓ (spatial evidence layer)	✓	✓ (OpEx-oriented)

Note: Symbols indicate feature support levels: ✓✓ = Full, native cloud-integrated capability; ✓ = Standard or partial support; ! = Restricted, manual, or highly constrained functionality; ✗ = Feature not supported.

This comparison highlights those existing solutions are typically either reconstruction-centric (photogrammetry tools) or documentation-centric (CDE/CMMS), but rarely unify 3D reconstruction, spatial anchoring of defects, and queryable evidence management in a single cloud-native workflow.

To ground this qualitative comparison, the following processing costs were considered: a 150-image project processed on a licensed desktop platform (e.g., Metashape Pro v2.1, with a commercial perpetual license cost of USD ~3500) incurs fixed upfront capital costs irrespective of project volume, versus approximately USD 0.80–1.20 per project under the proposed serverless model (Section 4.2). This order-of-magnitude reduction in baseline operational cost highlights the originality and accessibility of the serverless approach.

Existing studies have investigated UAV-based Digital Twins for defect localization, GeoBIM integration, and semantic enrichment, while other works have focused on photogrammetric reconstruction accuracy and cloud-based processing platforms. However, comparatively limited attention has been devoted to the cloud-native operationalization of inspection-oriented photogrammetric reconstruction under serverless execution constraints. As a result, challenges related to scalability, fault tolerance, spatial information integration, and infrastructure cost remain insufficiently addressed in existing workflows.

Many existing implementations rely on monolithic architectures or persistent virtual machines to handle resource-intensive Structure-from-Motion (SfM) and Multi-View Stereo (MVS) tasks. This centralized approach often leads to severe data fragmentation, processing bottlenecks, and high idle infrastructure costs, which limit the deployment of continuous monitoring platforms. Furthermore, the existing literature lacks comprehensive frameworks that seamlessly integrate stateless, serverless computational pipelines with precise spatial anchoring. To address these gaps, this paper proposes a fully decoupled cloud-native architecture that eliminates continuous infrastructure overhead while maintaining geometric and semantic integrity during 3D reconstruction.

Unlike pure computer vision studies that focus narrowly on defect detection algorithms, this work addresses the broader challenge of System Architecture and Information Management.

The primary contributions of this paper are:

1. Serverless reference architecture coupling timeout-resilient photogrammetric orchestration with spatial evidence management, validated on real UAV inspection datasets.
2. Formalized spatial anchoring model that binds defect records to georeferenced 3D mesh coordinates via raycasting, frustum-based image retrieval, and PostGIS spatial indexing (v3, OSGeo Foundation, Beaverton, OR, USA), transforming passive reconstructions into queryable evidence substrates.
3. Consumption-based economic model that converts capital-intensive photogrammetry infrastructure to pay-per-use Operational Expenditure, lowering the adoption barrier for SMEs in emerging markets.
4. Open-source release of the complete implementation as a reproducible reference architecture for construction informatics research.
5. An empirical demonstration that acquisition geometry, not image count, is the primary driver of MVS complexity in a serverless context, with practical implications for cloud job scheduling and resource allocation.

To guide the development and evaluation of this proposed architecture, this study addresses three primary research questions (RQs):

RQ1 (System Architecture and Scalability). *How can resource-intensive, stateful photogrammetric workloads (SfM/MVS) be effectively orchestrated within a stateless, serverless cloud environment while maintaining operational scalability and acceptable data-transfer overhead?*

RQ2 (Spatial Intelligence and Information Management). *How can disconnected inspection artifacts (e.g., 2D imagery, PDF reports) be bidirectionally anchored to a georeferenced 3D mesh using raycasting and frustum-based visibility analysis to ensure accurate, queryable spatial retrieval?*

RQ3 (Computational Complexity and Cloud Provisioning). *To what extent does UAV acquisition geometry, relative to image count, influence the computational workload of dense 3D reconstruction, and how can this relationship inform cloud resource allocation strategies?*

The remainder of this paper is organized as follows: Section 2 presents the materials and methods, encompassing the State-of-the-Art in photogrammetric reconstruction, BIM-based inspection, and cloud-native geospatial systems, alongside the complete serverless system architecture, the SfM/MVS pipeline design, and the mathematical formulation of the spatial anchor mechanism. Section 3 reports the experimental results, including reconstruction fidelity, pipeline stage performance, and spatial anchoring validation across two real UAV inspection datasets. Section 4 discusses the findings within the broader construction informatics literature, addressing architectural implications, economic transferability, and integration with existing BIM ecosystems. Section 5 draws conclusions and outlines future research directions toward automated defect detection and temporal change

monitoring. The underlying source code, and evaluation dataset are publicly hosted as Supplementary Materials.

2. Materials and Methods

This section details the technical implementation of the spatial intelligence hub, organized into three subsystems: the serverless reconstruction pipeline, the spatial anchoring mechanism, and the collaborative information management layer.

2.1. Architectural Overview and Design Principles

The system architecture adheres to three fundamental design principles that distinguish it from traditional desktop photogrammetry workflows:

Principle 1: Decoupling of Compute and State

Traditional photogrammetry applications tightly couple user interface, data management, and computational processing within monolithic desktop executables. This coupling creates resource contention where heavy reconstruction tasks monopolize workstation resources, preventing concurrent user interaction.

The proposed architecture strictly separates stateful components (user data, project metadata, camera poses) managed by lightweight web services from stateless computational tasks (SfM processing, mesh generation) executed in ephemeral containers. This separation enables the user interface to remain responsive during long-running reconstructions and allows computational resources to scale independently of user load.

Principle 2: Event-Driven Asynchrony with Durable Execution

Photogrammetric reconstruction inherently involves long-running processes that exceed typical HTTP timeout limits. Rather than maintaining persistent connections or implementing complex polling mechanisms, the proposed architecture employs an event-driven architecture where state transitions trigger asynchronous workflows.

Crucially, we utilize durable execution semantics where workflow progress is checkpointed, enabling automatic recovery from transient failures without restarting entire reconstruction pipelines [41]. This approach transforms the traditional "keep the workstation running overnight" workflow into "upload and receive notification when complete".

Principle 3: API-First Interoperability

To combat the data fragmentation problem identified in the Introduction, the system exposes all functionality through documented REST APIs. Geometric data is served via OGC-standard 3D Tiles, defect records are queryable through GeoJSON endpoints, and camera poses are accessible as structured JSON. This API-first design enables integration with enterprise asset management systems, GIS platforms, and custom analytical tools without vendor lock-in.

2.2. The Serverless Reconstruction Pipeline

The architecture, as shown in Figure 2, decouples image ingestion, durable photogrammetry orchestration, containerized SfM/MVS execution, object storage persistence, web-based collaborative 3D review, and PostGIS-backed spatial indexing for defect traceability and evidence linking. This specific combination addresses several architectural requirements:

"Next.js (v16, Vercel Inc., San Francisco, CA, USA) Server Components and Actions" provide a unified programming model where server-side logic can be collocated with client-side interface code, eliminating the need to maintain separate API server infrastructure for CRUD operations. This reduces architectural complexity while maintaining a clear separation of concerns through React (v19, Meta Platforms Inc., Menlo Park, CA,

USA) component-based architecture and the server–client execution model introduced in Next.js [42,43].

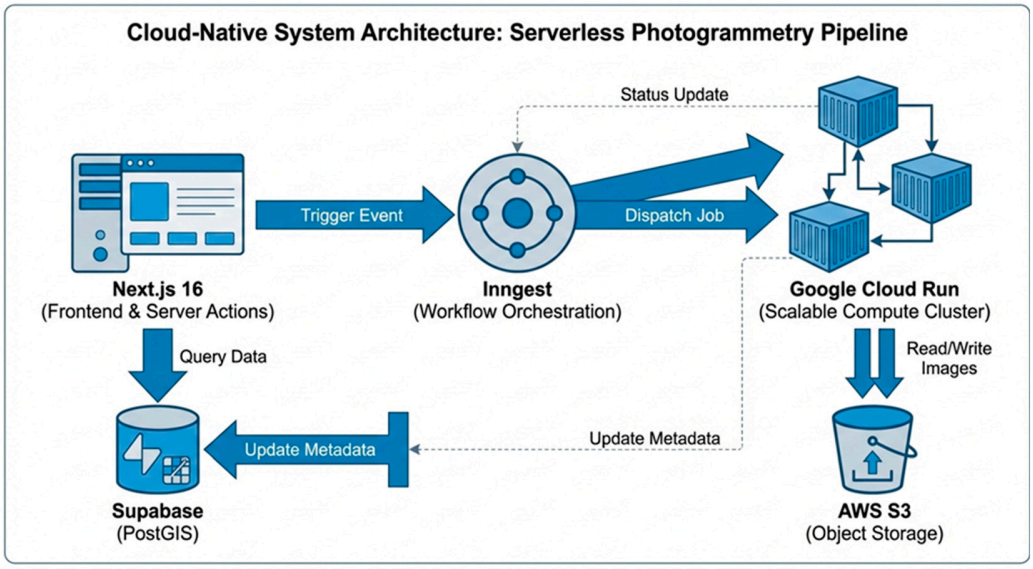

Figure 2. High-level cloud-native system architecture.

"Inngest (v3, Inngest Inc., San Francisco, CA, USA) durable execution model" solves the fundamental challenge of long-running photogrammetry in serverless contexts. Traditional serverless platforms impose strict timeout limits (typically 15 min for AWS Lambda [44], 60 min for Cloud Run [45]). A complete SfM reconstruction of 500+ images may require several hours. Inngest decomposes workflows into discrete steps with explicit dependencies, automatically handling retries, timeouts, and state persistence. Each step executes within timeout bounds, while the overall workflow can span arbitrary durations.

"Google Cloud Run (Gen 2, Google LLC, Mountain View, CA, USA)" provides serverless container execution with several advantages over function-based platforms. Containers encapsulate the complete toolchain including GDAL, PDAL, and native binaries, avoiding the "dependency packaging" challenges of AWS Lambda. Cloud Run's automatic scaling from zero to hundreds of concurrent instances enables handling simultaneous reconstruction requests without manual capacity planning. The pay-per-use billing model aligns costs directly with computational work performed [45].

"Supabase's (v2, Supabase Pte. Ltd., Singapore) PostGIS integration" transforms the PostgreSQL (v15, PostgreSQL Global Development Group, Berkeley, CA, USA) database into a spatial database management system capable of storing and indexing three-dimensional geometries. This is foundational to the spatial anchoring mechanism described in the spatial anchor mechanism section. Row-Level Security policies enforce multi-tenant data isolation at the database layer, providing defense-in-depth against application-level security vulnerabilities.

The reconstruction pipeline automates the complete Structure-from-Motion and Multi-View Stereo workflow through six orchestrated stages as illustrated in the end-to-end pipeline architecture of Figure 3:

The system ingests raw drone imagery and processes them asynchronously through checkpointed photogrammetry. Reconstruction outputs are published to a multi-user web 3D environment where defects are anchored on the mesh and linked to evidence retrieved via frustum-based visibility scoring, with persistence and querying handled by PostGIS.

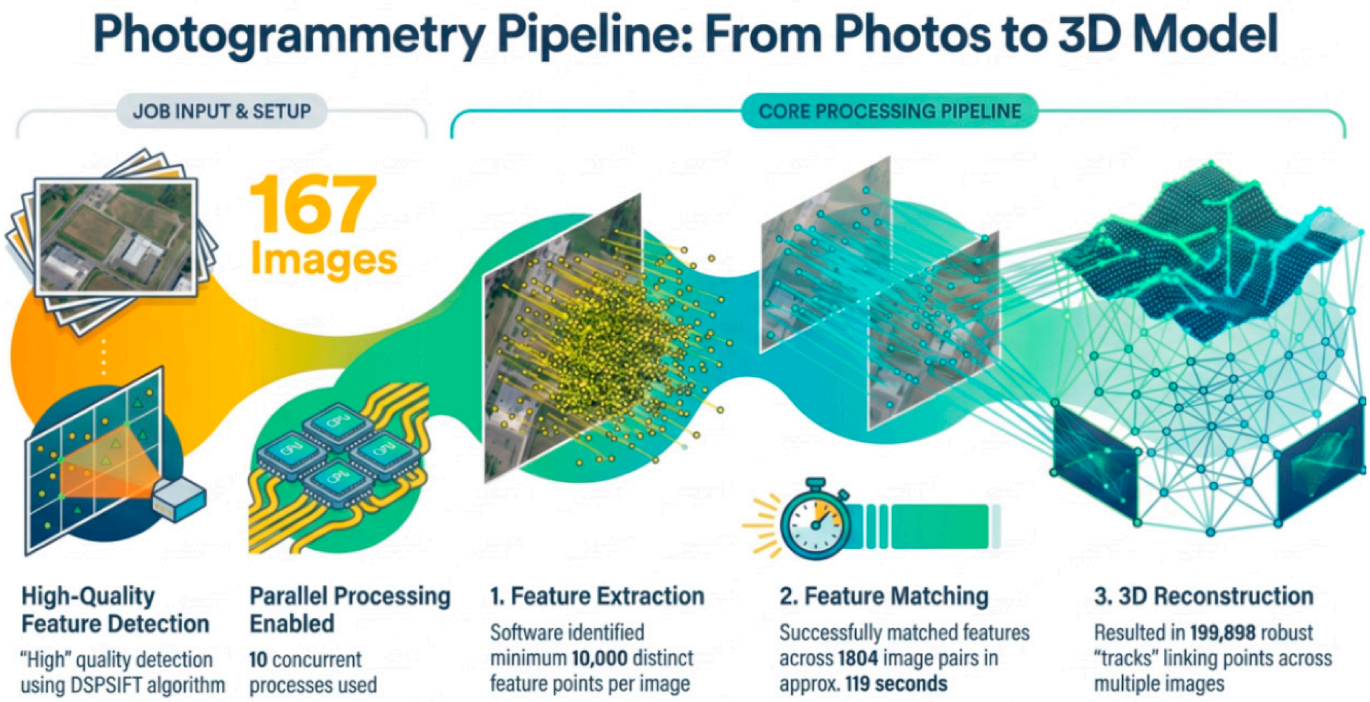

Figure 3. End-to-end photogrammetry pipeline.

Stage 1: Image Ingestion and Validation

Users upload drone imagery through the browser interface, which performs client-side validation checking for GPS EXIF tags and minimum image resolution requirements. Images are streamed directly to S3 using pre-signed URLs, bypassing application servers to avoid bandwidth bottlenecks. Upon the successful upload images, the uploaded event is published to Inngest's event bus.

Stage 2: Feature Extraction

The event triggers a Cloud Run container executing a feature extraction module. For each image, SIFT (Scale-Invariant Feature Transform) key points are detected and described, typically identifying 2000–10,000 features per image depending on scene texture [46]. Features are persisted to Google Cloud Storage as serialized descriptor files. Processing parallelizes across images with each container handling a batch of 10–20 images, enabling linear scaling with dataset size.

Stage 3: Feature Matching and Geometric Verification

A subsequent workflow step loads all feature descriptors and performs pairwise matching using an approximate nearest neighbor search (FLANN) [47]. Putative matches undergo geometric verification through RANSAC-estimated fundamental matrices, filtering spurious correspondences caused by repetitive textures or moving objects [48]. The output is a sparse graph where nodes represent images and edges represent verified geometric relationships with sufficient overlap for reconstruction [49].

Stage 4: Incremental Structure-from-Motion

The core SfM reconstruction employs incremental bundle adjustment. The algorithm selects an initial image pair with strong baseline separation, triangulates matched features to initialize a sparse 3D point cloud, then iteratively registers additional images by solving the Perspective-n-Point problem [27]. After each image registration, global bundle adjustment refines all camera parameters and 3D point locations simultaneously, minimizing

reprojection error across the entire dataset. This stage outputs camera-intrinsic (focal length, principal point, distortion coefficients), camera-extrinsic (position and orientation for each image), and a sparse point cloud. Critically, camera extrinsic parameters are serialized to a JSON format and stored in S3, enabling the spatial anchoring mechanism.

Stage 5: Multi-View Stereo Densification

The sparse reconstruction is densified through patch-based MVS [50]. Depth hypotheses are propagated and refined through iterative PatchMatch search, with each pixel storing a depth value and surface normal defining a slanted plane hypothesis. Multi-view photometric consistency, measured via normalized cross-correlation across neighboring views, drives hypothesis refinement. Depth maps are subsequently fused through geometric consistency filtering, rejecting estimates that are not corroborated across a minimum number of views. The output dense point cloud typically contains 10–100 million points depending on image resolution and scene complexity.

Stage 6: Mesh Generation and Web Optimization

The dense point cloud is converted to a textured mesh through Screened Poisson surface reconstruction, which treats meshing as solving a spatial indicator function whose gradient best approximates point normal [51]. The resulting watertight mesh undergoes decimation targeting 500k triangles for web rendering.

Texture generation employs view-dependent blending following the multi-view texturing approach of Waechter et al. [52], compositing image contributions weighted by viewing angle and distance. Finally, the mesh is tiled into an OGC 3D Tiles format with a hierarchical level-of-detail, enabling progressive streaming in CesiumJS (v1.115, Cesium GS, Inc., Philadelphia, PA, USA). The tileset is uploaded to S3 with public read access, and the project status is updated in Supabase, triggering a notification to the user.

The event-driven architecture provides robust failure handling. If feature matching fails due to insufficient image overlap, the workflow automatically retries with relaxed matching thresholds. If bundle adjustment fails to converge (often caused by poor camera calibration or extreme scene geometry), the system falls back to sequential SfM with incremental validation. Each failure mode is logged with diagnostic information enabling manual intervention if automated recovery fails. The durable execution model ensures that successful intermediate stages (e.g., feature extraction) are not discarded, avoiding redundant computation upon retry.

A recognized architectural trade-off of stateless serverless execution is the I/O overhead associated with persisting and reloading intermediate reconstruction artifacts, such as feature descriptors, sparse point clouds, and depth maps, between pipeline stages. Because each stage executes within an ephemeral container, the intermediate state cannot be retained across stage boundaries and must therefore be externalized to object storage.

Several design decisions mitigate this overhead. First, intermediate artifacts are stored in compact binary formats, including compressed numerical arrays, reducing storage requirements compared with verbose text-based representations. Second, object storage access is restricted to stage boundaries, whereby each processing stage performs a single read operation at startup and a single write operation upon completion, avoiding repeated transfers during execution. Third, compute services and object storage are deployed within the same cloud region, minimizing network latency and eliminating cross-region transfer costs.

Although this design introduces additional data movement relative to monolithic architectures, it enables fault-tolerant checkpointing, independent stage recovery, and elastic scaling. For the inspection-oriented workloads evaluated in this study, the resulting I/O overhead remained operationally acceptable and did not offset the economic and deployment advantages of the serverless architecture.

2.3. The Spatial Anchor Mechanism

The spatial anchor mechanism creates bidirectional linkage between database records (defect annotations, test reports) and geometric locations on the reconstructed 3D mesh. This section formalizes the coordinate transformation pipeline and database schema supporting this linkage.

Mathematical Formulation of Raycasting

When a user clicks location (u, v) in the browser-based 3D viewer, the system must determine which 3D point on the mesh surface was clicked. This is accomplished through raycasting combined with GPU depth buffer interrogation, following the geometric principles illustrated in Figure 4.

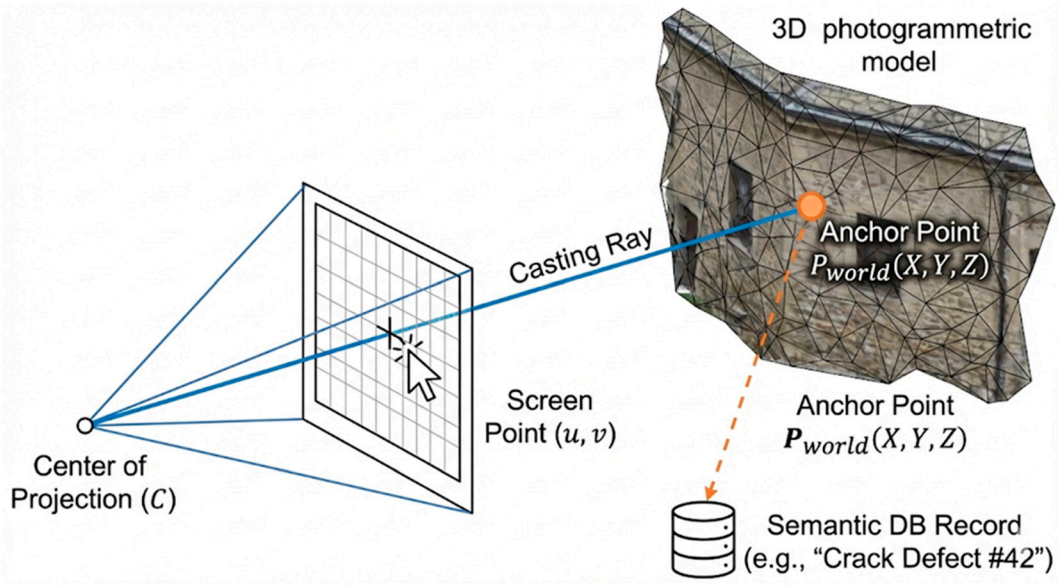

Figure 4. Geometric principles of spatial anchoring via raycasting.

The camera model is defined by its center $\mathbb{C} \in \mathbb{R}^3$ expressed in the object coordinate frame and its orientation which is represented by a rotation matrix $\mathbb{R} \in SO(3)$. The perspective projection is characterized by an intrinsic matrix:

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

where f_x, f_y are focal lengths in pixel units and (c_x, c_y) is the principal point.

Given the click coordinate $u = (u, v)$ in screen space (measured in pixels from top-left origin), we seek the corresponding surface point $p_w \in \mathbb{R}^3$.

Instead of explicit ray–triangle intersection, we recover p_w directly from the GPU depth buffer at $\boldsymbol{u}$ using the inverse view-projection transformation. This yields a coordinate P_{ECEF} on the visible mesh surface, which is subsequently converted to geodetic coordinates (ϕ, λ, h) using the iterative Bowring method to ensure sub-millimeter precision for database storage [53].

Camera Pose Storage and Frustum Culling

The SfM reconstruction outputs camera-extrinsic parameters for each source image, stored as a JSON object in S3.

When a user clicks a location on the 3D mesh to retrieve relevant source images, the system performs frustum culling to identify which cameras have that point within their field of view. Because the 3D mesh and the camera poses have already been co-registered into the same metric object space during the georeferencing stage, the scale factor is unity $(\mu = 1)$. Therefore, the transformation is a pure rigid body motion.

For a candidate camera i with rotation matrix R_i and translation vector t_i, we transform the anchor point p_w into the camera coordinate system:

$$p_{c,i} = R_i p_w + t_i \tag{2}$$

The point is projected into Normalized Device Coordinates (NDC) using the projection matrix P.

The point is considered contained within the frustum F_i if and only if:

$$-1 \le x_{ndc} \le 1,\ -1 \le y_{ndc} \le 1,\ 0 \le z_{ndc} \le 1$$

We rank the surviving candidates using a composite score s_i that balances incidence angle and distance, as geometrically conceptualized in Figure 5. To prioritize high-resolution views while penalizing grazing angles, we employ the following scoring function:

$$s_i = \alpha \cdot max(0, n \cdot v_i) + (1 - \alpha) \cdot exp\left(-\frac{\|c_i - p_w\|}{\sigma}\right) \tag{3}$$

where n is the surface normal at the anchor, v_i is the viewing direction, c_i is the camera center, and $\alpha \in [0,\ 1]$ is a tunable weight factor.

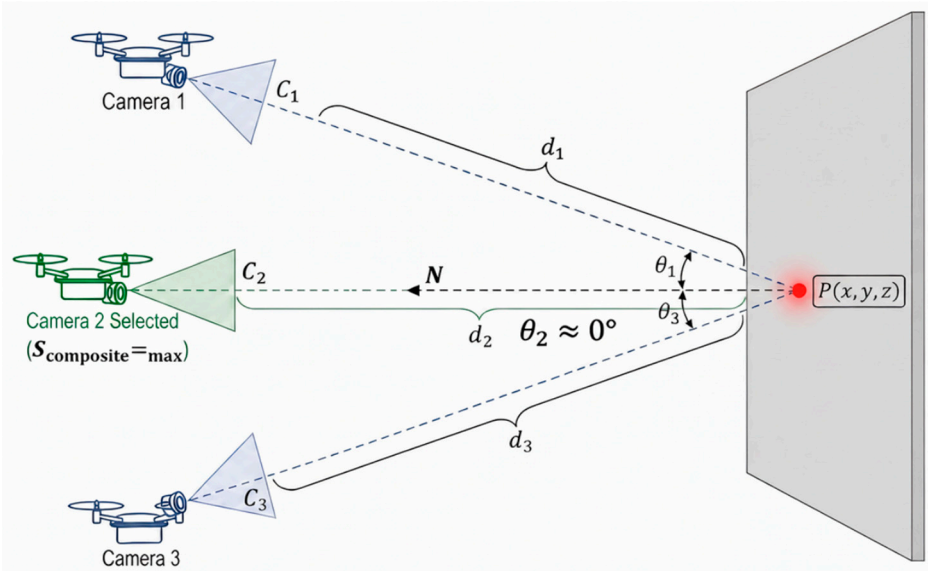

Figure 5. Geometric scoring criteria used in Algorithm 1 for image retrieval. After frustum containment and occlusion filtering, candidate cameras are ranked using a composite score based on the Euclidean distance d_i between camera center $\mathbb{C}_i$ and anchor point P, and the incidence angle θ_i between the viewing direction and local surface normal N. Camera $\mathbb{C}_2$ achieves the highest composite score and is therefore selected as the optimal inspection image.

The image with the highest score is retrieved and displayed to the user, providing the optimal source image for inspecting the clicked location. The complete retrieval logic, incorporating the geometric checks and scoring function defined above, is formalized in Algorithm 1.

Algorithm 1. Frustum-based image retrieval for a 3D anchor point

Input:
Anchor point $\boldsymbol{p}$ on the 3D mesh
Set of cameras C = $\{\mathbf{c_i}\}$, with known poses and intrinsics
Integer k, number of images to return
Output:
Ranked list of top-k inspection images I_k
Procedure:
Initialize an empty list of candidate images S.
For each camera c_i in C:
Project point $\boldsymbol{p}$ into the camera's local coordinate system. If the point lies behind the camera or outside the 2D image boundaries, discard the camera and continue.
Perform a raycast from the camera center to p. If the ray intersects the mesh before reaching $\boldsymbol{p}$, discard the camera (point is occluded).
Calculate the composite score $\boldsymbol{S_{composite}}$ based on the camera's Euclidean distance to $\boldsymbol{p}$ and the incidence angle against the surface normal.
Append the camera and its score to $\boldsymbol{S}$.
Sort S in descending order of score.
Return the indices of the top-k images from S as I_k.

Database Schema for Spatial Anchoring

Defect records and associated documentation are stored in PostGIS-enabled PostgreSQL tables with spatial geometry columns. Anchor coordinates are converted from the metric object–space coordinate frame used during raycasting to WGS84 geodetic coordinates via the iterative Bowring method prior to database insertion, enabling cross-project portfolio queries using standard geographic predicates.

The GEOMETRY (POINTZ, 4326) type persists these three-dimensional coordinates (longitude, latitude, ellipsoidal height) referenced to the WGS84 datum (SRID 4326). The GIST spatial index enables efficient proximity queries, and the geography cast enables metric distance calculations accounting for Earth's curvature, essential for accurate spatial queries across building portfolios.

User Workflow: From Click to Annotation

The complete user workflow integrates geometric visualization, image retrieval, and database persistence, as illustrated by the interface components in Figure 6:

Inspector explores the 3D model in the browser, rotating and zooming to examine different building facades.

Suspicious location identified. Upon noticing surface damage, the inspector clicks the location. The raycasting algorithm extracts world coordinates.

Source image retrieval. Frustum culling identifies the optimal source image. The high-resolution image is fetched from S3 and displayed with the clicked location highlighted.

Defect annotation created. The inspector completes a form specifying the defect type, severity, and textual description. This triggers an INSERT operation into the defects table with the clicked coordinates stored in the location geometry column.

Supporting documentation attached. The inspector uploads test reports (e.g., concrete core analysis, ultrasonic thickness measurements) as PDF files. These documents are linked to the defect record and stored with the same spatial anchor.

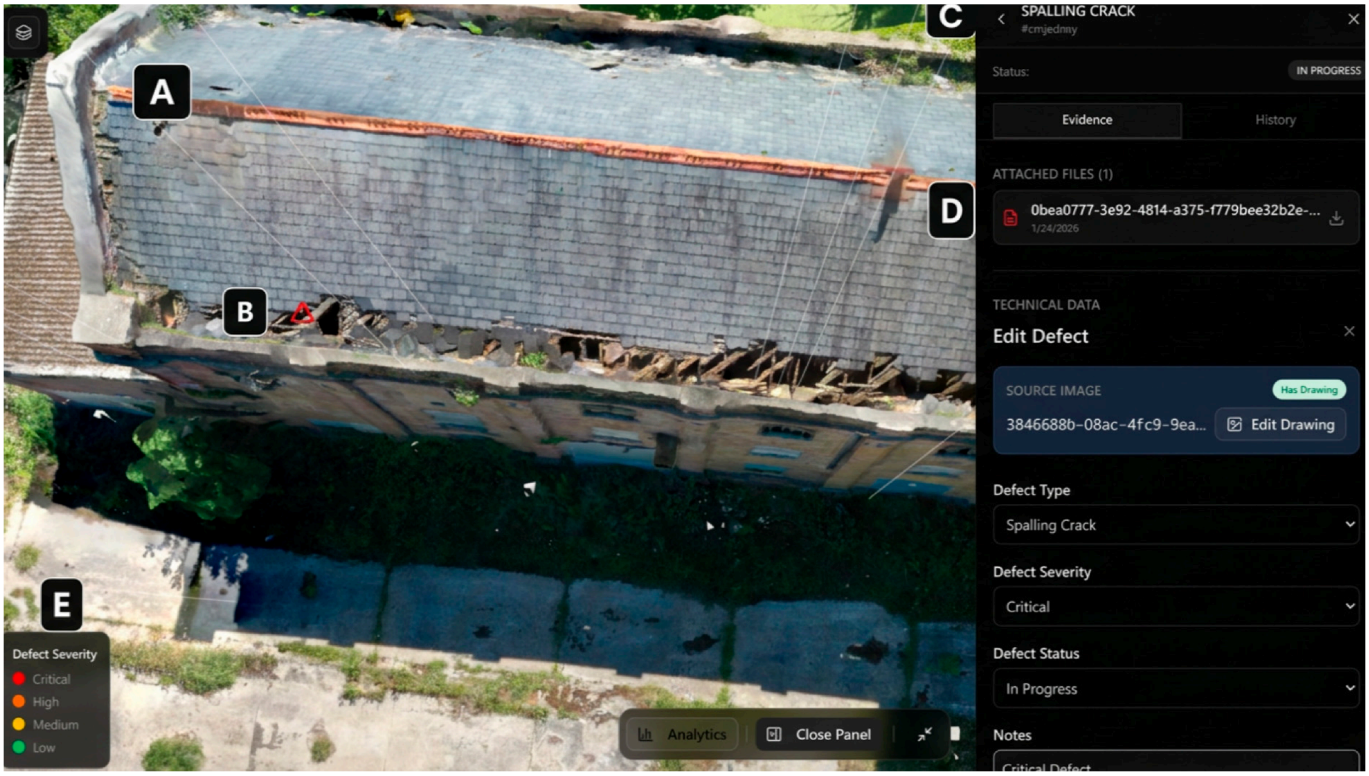

Figure 6. Spatially anchored defect annotation workflow within the proposed inspection platform: (**A**) High-resolution 3D mesh visualization of the target infrastructure; (**B**) Spatially anchored defect marker (red triangle) located directly on the structural damage; (**C**) Defect identification header and inspection status; (**D**) Technical data panel for managing source imagery, defect type, severity, and notes; (**E**) Color-coded defect severity legend.

Portfolio-wide queries enabled. The defect record becomes immediately queryable across the entire building portfolio, supporting questions like "show all critical corrosion within 3 m of expansion joints in all buildings constructed before 2000.

2.4. Multi-Tenancy and Collaborative Features

The platform implements B2B SaaS multi-tenancy through Supabase Row-Level Security (RLS) policies. Each organization exists as a tenant with isolated data. Role-Based Access Control (RBAC) distinguishes between administrators (full CRUD on all resources), project managers (create/update projects and defects), inspectors (create defects, upload documents), and viewers (read-only access). These roles are enforced through additional RLS policies checking user-organization-role mappings.

Real-time collaboration leverages Supabase's built-in WebSocket pub/sub. When one user creates a defect annotation, all other users currently viewing the same project receive a real-time notification, and their 3D viewers automatically update to display the new defect marker. This enables scenarios where a field inspector and remote structural engineer jointly review an inspection, with both seeing each other's annotations appear in real time.

The quantitative evaluation of the real-time collaboration layer is outside the scope of this study. Based on Supabase's WebSocket pub/sub architecture, notification delivery latency is expected to be under 200 ms for standard LAN/4G connections, consistent with published benchmarks for similar WebSocket pub/sub systems, but this was not measured under concurrent-user load in the current evaluation. A dedicated multi-user stress test simulating 10–20 concurrent inspectors creating annotations simultaneously was identified as necessary validation before enterprise deployment, particularly to characterize conflict

resolution behavior when two users anchor a defect at the same mesh location within the same time window.

3. Results

3.1. Experimental Setup

To validate the performance and feasibility of the proposed serverless architecture, we conducted experiments using two distinct datasets representing typical infrastructure inspection scenarios.

Dataset A comprises 167 UAV images of The Mill building in London, acquired using a mapping-oriented flight configuration combining predominantly nadir imagery with a single oblique pass. The building is characterized by a pitched roof with two dominant planar slopes intersecting along a ridge line. Images were captured using a Hasselblad L1D-20c camera (DJI, Shenzhen, China.) at approximately 21.3 m altitude with 80% forward overlap and 80% lateral overlap, yielding a Ground Sampling Distance (GSD) of approximately 0.97 cm/pixel.

This nadir-dominated viewing geometry provides strong intersection angles for horizontal roof surfaces but produces weak network configuration for vertical façade elements, reflecting a common limitation in urban aerial survey workflows where façades are not the primary acquisition target.

Dataset B consists of 85 images of a building roof structure at EPFL (Swiss Federal Institute of Technology, Lausanne, Switzerland), acquired at an altitude of 30 m AGL by a Phantom FC330 drone (DJI, Shenzhen, China.) using a circular oblique survey pattern, providing a Ground Sampling Distance (GSD)of approximately 1.67 cm/px. Despite the coarser spatial resolution compared to Dataset A, the oblique convergent viewing geometry creates diverse triangulation angles and high baseline-to-depth ratios across the image network. The scene contains photovoltaic panels, skylights, chimneys, and service installations that introduce significant local depth variation and rich, non-repetitive texture.

The selection of two geometrically distinct datasets was deliberate rather than opportunistic, as illustrated in the survey flight configurations of Figure 7. Dataset A and Dataset B were chosen to represent opposite ends of the acquisition geometry spectrum encountered in practice: nadir-dominated grid surveys, which dominate commercial inspection workflows due to their simplicity, and circular oblique surveys, which represent best-practice photogrammetric network design for complex rooftop structures. This geometric contrast allows the evaluation to probe the system's behavior under conditions where the primary driver of computational complexity as established in photogrammetric literature shifts from image count to network geometry [38–40]. A homogeneous sample of similarly acquired datasets would obscure precisely the architectural hypothesis being evaluated, namely that checkpoint-based orchestration is necessary because MVS complexity is geometry-driven and therefore unpredictable from input metadata alone. The two-dataset design thus constitutes a controlled contrast rather than a statistical sample, consistent with the paper's goal of demonstrating architectural feasibility and identifying the dominant complexity driver rather than establishing population-level performance bounds.

This configuration generates high parallax and strong multi-view intersection geometry, making the dataset suitable for evaluating fine-scale surface reconstruction under optimal network conditions. Both datasets were processed without ground control points; camera positions were initialized from GPS EXIF metadata embedded at capture.

The reconstruction pipeline was deployed on the Google Cloud Platform with the following configuration:

- Frontend/API: Next.js application on serverless edge network.
- Database: Supabase PostgreSQL with PostGIS.

- Orchestration: Inngest serverless event bus for state management.
- Object Storage: Amazon S3 (Standard Tier, Amazon Web Services Inc., Seattle, WA, USA).
- Compute: Google Cloud Run (Generation 2) with dynamic instance scaling (max 100 instances).
- Container Spec: 4 vCPUs, 16 GB RAM per instance.

Figure 7. Survey datasets used for experimental evaluation. (**a**) Dataset A: 167 images acquired at 21.3 m AGL using a nadir-dominated grid survey with a single oblique pass (GSD ≈ 0.97 cm/px). (**b**) Dataset B: 85 images acquired at 30 m AGL using a circular oblique survey pattern (GSD ≈ 1.67 cm/px).

3.2. Reconstruction Fidelity and Density

The reconstruction results reveal a counter-intuitive relationship between acquisition parameters and output density that has direct implications for cloud resource provisioning. Despite having fewer images (85 vs. 167) and coarser Ground Sampling Distance (GSD 1.67 cm/px vs. 0.97 cm/px), Dataset B produced a denser point cloud than Dataset A both in absolute terms (8.73 M vs. 8.33 M points) and per image (102,000 vs. 49,000 points/image, a 2.1× difference).

This inversion is consistent with the established photogrammetric theory on oblique network geometry, it directly motivates the checkpoint-based orchestration design. As shown in Table 2, the serverless pipeline successfully adapted to the nature of the input data.

Table 2. Reconstruction metrics outcomes for Datasets A and B.

Metric	Dataset A (167 Imgs)	Dataset B (85 Imgs)
Sparse Cloud	199,898 points	84,691 points
Dense Point Cloud	8.33 million	8.73 million
Points per Image	~49,000	~102,000
Peak RAM Usage	5.2 GB	5.6 GB

The results highlight a critical architectural advantage: resource consumption correlates with point density, not just image count, as quantified across the key output metrics in Figure 8.

- Dataset A (More images, lower density) required 49 min.
- Dataset B, despite containing fewer images, generated 8.73 million points—exceeding the total reconstruction density of Dataset A—and required 45 min to complete the reconstruction pipeline.This demonstrates that the "Dispatcher" architecture effectively

scales resources to match the complexity of the reconstruction (Multi-View Stereo load), rather than linearly scaling with the number of input files.

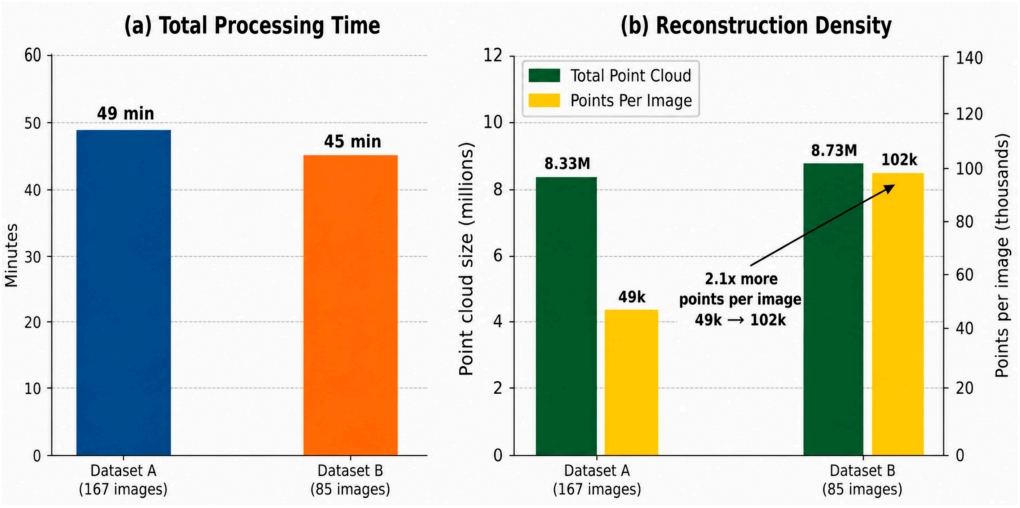

Figure 8. Reconstruction output metrics for both datasets: (**a**) Total end-to-end processing time remains comparable despite differing dataset characteristics (49 min vs. 45 min), (**b**) Dataset B generates a denser point cloud both in total (8.73 M vs. 8.33 M points) and per image (102,000 vs. 49,000, a 2.1× increase).

3.3. Pipeline Stage Performance

To understand the scaling behavior of the serverless architecture, we profiled wall-clock execution time for each pipeline stage. Table 3 presents the complete breakdown for both datasets.

Table 3. Processing time distribution by pipeline stage.

Pipeline Stage	Dataset A (167 Imgs)	Dataset B (85 Imgs)	Interpretation
Feature Extraction (parallel)	5 min 25 s	3 min 00 s	Linear with image count
Feature Matching	3 min 00 s	1 min 35 s	Sub-linear; GPS filtering reduces pair count
Structure-from-Motion	8 min 22 s	3 min 21 s	Sequential; scales with image count
Dense Reconstruction (MVS)	18 min 05 s	25 min 05 s	Geometry-driven complexity dominates
Point Filtering	22 s	18 s	Statistical outlier removal
Meshing	6 min 36 s	3 min 42 s	Poisson reconstruction
Texturing	2 min 08 s	2 min 25 s	View-dependent blending
Georeferencing & Tiling	1 min 52 s	2 min 28 s	Octree generation
Total Duration	49 min 29 s	45 min 24 s	Non-linear result

Note: Total end-to-end wall-clock processing times include an additional ~3.5 min of cloud VM provisioning overhead not reflected in the core algorithmic stages.

The most significant finding is visible in Stage 4 (Dense Reconstruction). Dataset B, despite containing half the images of Dataset A, required 38% more MVS time, as illustrated in the wall-clock execution breakdown in Figure 9. This inversion is driven by geometric complexity: the oblique acquisition in Dataset B produced 102,000 points per image compared to Dataset A's 49,000 points per image (Table 2).

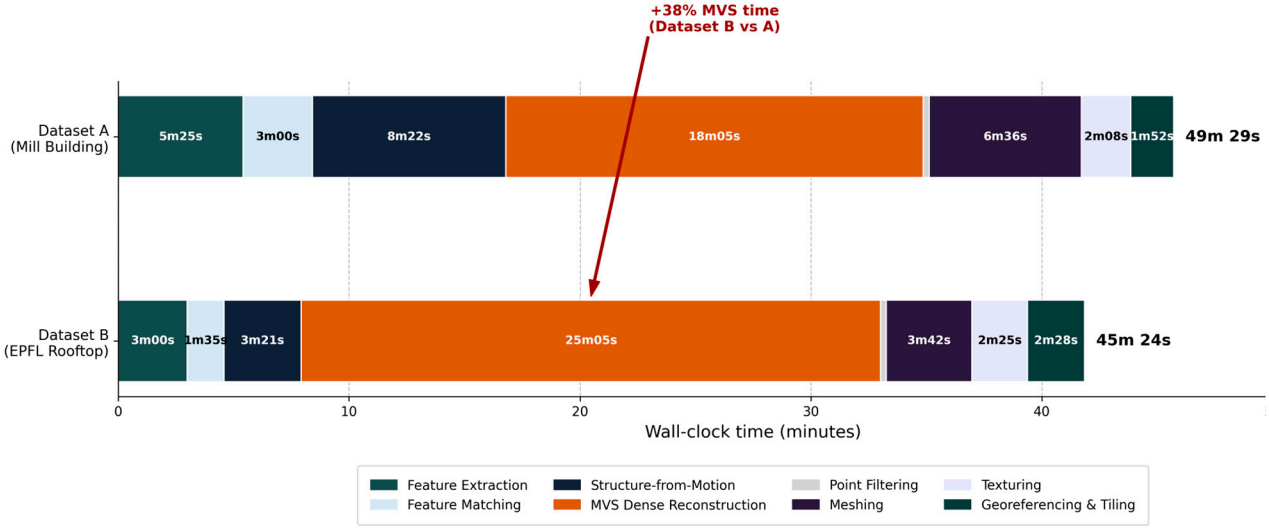

Figure 9. Wall-clock time distribution by pipeline stage. MVS dense reconstruction dominates total runtime and scales with scene geometry, not image count: Dataset B (85 images) requires 38% more MVS time than Dataset A (167 images) despite having fewer images and coarser Ground Sampling Distance, driven by its oblique convergent network geometry.

The higher parallax and denser stereo matching per pixel translated directly into MVS computational load, confirming that scene geometry, not file count, dictates reconstruction runtime.

This validates the architectural hypothesis that motivated the checkpoint-based orchestration: MVS duration is unpredictable from image count alone. A naive serverless deployment allocating resources based on N images would under-provision Dataset B and over-provision Dataset A. The durable execution design avoids this failure mode by decomposing the pipeline at stage boundaries and allowing each stage to provision resources dynamically based on actual workload.

The stage-by-stage profiling (Table 3) reveals two distinct scaling regimes. Feature extraction and matching (Stage 1, Stage 2) exhibited the expected linear relationship to image count, with Dataset A requiring 08 min 25 s and Dataset B requiring 04 min 35 s. This confirms that SIFT feature detection at approximately 10,000 keypoints per image scales predictably with N.

Dense reconstruction (Stage 4) inverted this relationship. Dataset B's 25 min 05 s MVS runtime exceeded Dataset A's 18 min 05 s despite containing half the images. The cause is visible in the output metrics: Dataset B generated 8.73 million points (102,000 points/image) compared to Dataset A's 8.33 million points (49,000 points/image), as visually detailed in the point cloud and mesh outputs of Figure 10.

The computational cost of PatchMatch stereo matching scales with the number of depth hypotheses tested per pixel, which increases with baseline-to-depth ratio and image overlap. Dataset B's circular oblique geometry at 30 m AGL with convergent viewing angles maximized both factors despite its coarser GSD, confirming the mechanism established in the photogrammetric literature viewing angle diversity and network convergence, not image resolution, govern the computational cost of dense stereo matching, producing the observed 38% MVS time increase.

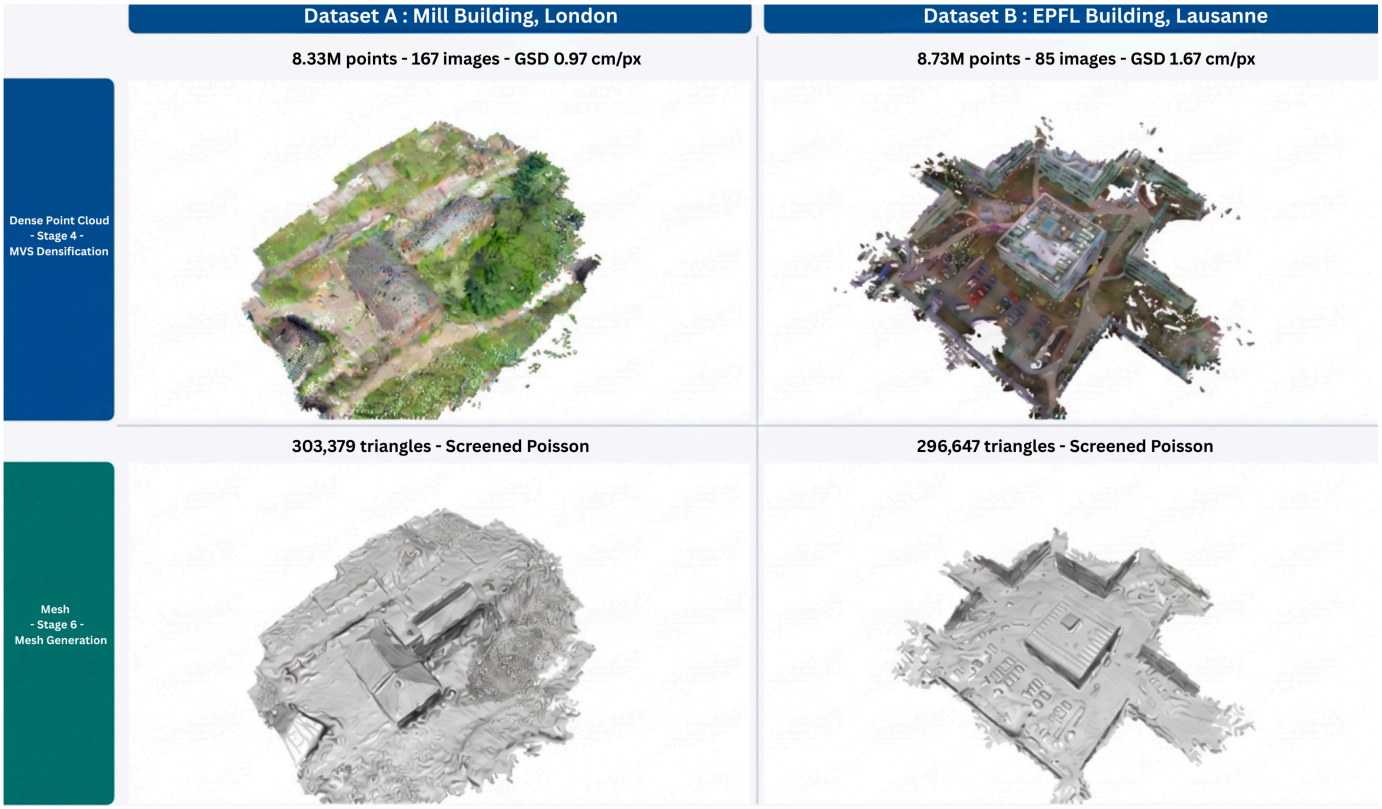

Figure 10. Pipeline reconstruction outputs for both datasets. Top row: Stage 4 dense point cloud (PatchMatch MVS). Dataset B (85 images, GSD 1.67 cm/px) produces 8.73 M points versus 8.33 M for Dataset A (167 images, GSD 0.97 cm/px), consistent with the 2.1× per-image density advantage driven by the oblique convergent survey geometry. Bottom row: Stage 6 mesh after Screened Poisson surface reconstruction. Dataset A (303,379 triangles) shows characteristic vertical streaking artifacts on the brick facade from the nadir-dominated image network. Dataset B (296,647 triangles) captures the complex EPFL rooftop with its photovoltaic panels, skylights, chimneys, and service installations, reflecting the rich depth variation that the oblique circular survey was designed to resolve.

Post-processing stages (filtering, meshing, texturing, tiling) scaled approximately with the final point cloud size rather than input image count. The peak memory usage of 5.6 GB for Dataset B and 5.2 GB for Dataset A remained well within the 16 GB container limit, demonstrating that CPU-only serverless instances can handle high-fidelity centimeter-scale inspection reconstructions without GPU acceleration.

The observed runtime differences were consistent with the influence of acquisition geometry. PatchMatch-based Multi-View Stereo estimates depth by evaluating geometric consistency across neighboring views. In convergent oblique image networks, individual surface points are observed from a wider range of viewing angles and often participate in a larger number of geometrically meaningful view combinations. This increases the computational effort required to evaluate and refine depth hypotheses. By contrast, nadir-dominated acquisition patterns typically exhibit lower angular diversity, reducing the number of candidate view relationships contributing to depth estimation. The results observed in Dataset B are consistent with this interpretation: despite containing fewer images, the dataset produced higher reconstruction density and longer MVS processing times than Dataset A. These findings suggest that acquisition geometry may serve as a useful predictor of cloud reconstruction workload and should be considered alongside image count when estimating resource requirements.

3.4. Spatial Anchoring Validation

3.4.1. Image Retrieval Accuracy

Manual functional evaluation was conducted across 30 randomly sampled anchor points distributed to ensure spatial coverage: 12 points on vertical façade surfaces, 11 points on roof planes, and seven points at geometrically complex zones including building corners, dormers, and roof-to-wall transitions. For each anchor, the top-ranked image returned by the frustum-based scoring mechanism was assessed against two criteria: (1) the anchor point must be visible within the image frame without occlusion, and (2) the viewing angle must be sufficient to resolve surface texture at a scale appropriate for defect inspection. The mechanism demonstrated correct retrieval in 28 of the 30 cases, corresponding to a top-1 accuracy of 93.3%. For a binomial proportion of 28 successes in 30 trials, the 95% Wilson confidence interval was [77.9%, 98.8%], reflecting the wide uncertainty inherent in a 30-sample evaluation. This interval should be interpreted as indicative rather than definitive; a minimum sample of 100–150 anchor points across diverse building typologies would be required to bound top-1 accuracy within ± 5 percentage points at 95% confidence. The current evaluation is therefore sufficient to demonstrate functional correctness of the mechanism but insufficient to establish operational performance guarantees.

The two failure cases occurred exclusively at building corners where grazing viewing angles caused the highest-scored image to capture the anchor point at the extreme image periphery, satisfying the frustum containment condition geometrically while providing insufficient surface resolution for inspection purposes. This failure mode is predictable from first principles: the composite scoring function penalizes large incidence angles via the normal-direction dot product term $n \cdot v_i$, but does not explicitly penalize high eccentricity within the image frame. A straightforward mitigation adding an image-plane centrality term to the scoring function is identified as a targeted improvement for future work. Critically, both failure cases were concentrated at a single geometric category (exterior corners) that represents a known limitation of nadir-dominated acquisition geometry rather than a deficiency of the retrieval mechanism itself; re-acquisition with a targeted oblique pass at these locations would eliminate the failure mode at the data collection stage. These results confirm the functional correctness of the frustum-based approach across the dominant surface categories encountered in building inspection.

3.4.2. Spatial Query Performance

We populated the defects table with 500 synthetic test records distributed across the reconstructed building. Spatial proximity queries using PostGIS's ST_DWithin function executed in 15–25 ms with the GIST spatial index, compared to 800–1200 ms without indexing (sequential scan). This confirms that the spatial indexing layer supports interactive query response times across large defect inventories, validating the architecture's suitability for portfolio-wide spatial queries within an operational inspection interface.

To contextualize these results operationally, sub-25 ms query latency places spatial defect retrieval within the same response-time budget as standard database lookups in enterprise asset management systems, which typically target under 100 ms for interactive workflows [12]. The 40–50$\times$ speedup provided by GIST spatial indexing over sequential scan is consistent with theoretical expectations for spatial index structures on point geometries, where query complexity reduces from O(n) to O(log n) [24]. More significantly, the measured latency was achieved on a dataset of 500 records distributed across a single building footprint. For portfolio-scale deployments spanning hundreds of buildings with tens of thousands of defect records, query performance will depend on the spatial distribution of anchors relative to the query geometry and the selectivity of the ST_DWithin radius parameter. The PostGIS geography cast used in the implementation ensures that

distance calculations account for Earth's curvature, maintaining metric accuracy for cross-building queries spanning geographic distances where planar approximations introduce unacceptable error. These characteristics establish that the spatial indexing layer is architecturally appropriate for operational deployment at the building-portfolio scale targeted by the platform.

The 500-record evaluation confirms interactive latency for a single-building deployment. For portfolio-scale scenarios, 200 buildings, each with 500–1000 defect records, totaling 100,000–200,000 spatial records, query performance will depend on the selectivity of the ST_DWithin radius and the geographic extent of the portfolio. At this scale, standard GIST indexing on a single PostgreSQL instance is anticipated to maintain sub-100 ms latency for well-parameterized proximity queries, based on published PostGIS benchmarks for point geometry datasets of comparable size. For portfolios exceeding ~1 million records or spanning multiple geographic regions, partitioning strategies, either range-based partitioning by building identifier or geographic partitioning by bounding box, would be required to prevent full-index scans. The current single-instance Supabase deployment is appropriate for the pilot-scale use cases targeted in this work; distributed scaling via read replicas or Citus-based sharding represents a natural extension for enterprise deployment.

4. Discussion

4.1. Architectural Contribution to Construction Informatics

This work demonstrates that thoughtful system architecture addressing workflow integration can deliver substantial practical value independent of algorithmic sophistication. While much academic attention focuses on improving defect detection accuracy or reconstruction fidelity, practitioners struggle with more fundamental challenges around data organization, multi-stakeholder collaboration, and portfolio-scale visibility. The spatial intelligence hub addresses these practical needs through three architectural patterns:

Event-Driven Decomposition for Heavy Compute. The photogrammetry pipeline demonstrates that computationally intensive geospatial workflows can be successfully disaggregated into event-triggered microservices without sacrificing reliability. This pattern is generalizable beyond photogrammetry to other construction monitoring applications including BIM model synchronization, clash detection, and energy simulation.

Geometry as Database Interface. By treating the 3D reconstruction as a spatial organizing substrate rather than passive visualization, we enable a fundamentally different interaction paradigm. Users navigate spatial reality (the 3D model) rather than abstract file structure, and the system automatically maintains bidirectional linkage between geometric locations and business records. This pattern could be applied to other asset management domains including MEP system maintenance and historic preservation.

While the proposed architecture relies on a traditional Structure-from-Motion (SfM) and Multi-View Stereo (MVS) pipeline to generate this geometric interface, a theoretical comparison with emerging view-synthesis paradigms, such as Neural Radiance Fields (NeRF) and 3D Gaussian Splatting (3DGS), highlights critical operational trade-offs. NeRF and 3DGS have gained significant academic attention for their ability to render highly photorealistic, continuous representations of complex scenes from sparse imagery. However, these implicit and explicit radiance field methods are optimized primarily for visual fidelity rather than discrete geometric engineering metrics. For infrastructure inspection applications requiring explicit metric geometry, traditional MVS remains better aligned with the requirements of the proposed spatial anchoring workflow because it reconstructs explicit, geometrically verifiable polygonal meshes and point clouds with a true metric scale.

This explicit geometry is mathematically required for raycasting-based coordinate extraction and PostGIS spatial indexing. Furthermore, training NeRFs or optimizing

millions of 3D Gaussians introduces immense computational overhead, often requiring high-end, persistent GPU infrastructure that directly conflicts with the low-cost, stateless, CPU-friendly serverless economics demonstrated on this platform. Consequently, while view-synthesis methods excel at visual walkthroughs, the SfM/MVS paradigm remains the essential mathematical substrate for queryable, spatially anchored asset intelligence.

Serverless Democratization of Specialized Tools. The economic analysis demonstrates that converting capital-intensive desktop applications into serverless cloud services can reduce adoption barriers by an order of magnitude. This democratization is particularly impactful in emerging markets where SMEs lack capital for high-end workstations but can afford modest per-project processing fees. This pattern represents a potential strategy for digitally transforming other AEC workflows that currently require expensive specialized software.

Acquisition Geometry as a Cloud Provisioning Signal. The comparative reconstruction experiment surfaces an implication for serverless system design. While it is well established in photogrammetry that MVS complexity scales with network geometry rather than image count, the finding is that oblique convergent surveys produce denser, more computationally demanding reconstructions than nadir-dominated surveys of an equivalent or larger size. Future work should investigate lightweight geometric complexity estimators, derivable from image network topology prior to execution that would enable dynamic container provisioning without a pre-processing run.

The empirical results reported in Section 3 provide concrete grounding for several claims that have been asserted but not previously measured in the context of serverless cloud deployment. Remondino et al. [39] established theoretically that dense image matching complexity scales with viewing angle diversity and baseline-to-depth ratio rather than image resolution, and Haala and Rothermel [38] demonstrated this effect experimentally in batch desktop processing. The results presented here extend these findings to a serverless, pay-per-use execution context where the cost implications are direct and immediate: Dataset B required 38% more MVS time despite containing 49% fewer images, a cost inversion that a naive image-count-based resource allocation model would mishandle in both directions simultaneously, over-provisioning Dataset A and under-provisioning Dataset B.

This finding has a concrete implication for cloud cost management that has not been previously articulated: acquisition geometry metadata, specifically whether a survey uses nadir-dominated or oblique convergent flight patterns, should be treated as a first-class input to serverless job scheduling, not merely as a photogrammetric quality parameter. The checkpoint-based orchestration design adopted in this work is a pragmatic response to this uncertainty; future work on lightweight geometry complexity estimators, as proposed in Section 5, represents a path toward proactive rather than reactive resource allocation.

4.2. Technological Leapfrogging and Economic De-Risking

The results give some empirical grounding to the leapfrogging argument made in the introduction. In the traditional workstation paradigm, the requirement to purchase high-performance hardware creates a "sunk cost" barrier: organizations must commit capital ex-ante, often before securing the project contracts that would justify the investment. This creates a chicken-and-egg problem for Small and Medium Enterprises (SMEs), particularly in emerging markets like Morocco.

For a firm in Casablanca or Fez, a high-end workstation represents a significant percentage of an SME's discretionary budget, requiring executive approval and multi-year depreciation schedules. By transitioning to a serverless, consumption-based model, the proposed platform resolves this paradox. It converts a high-barrier Capital Expenditure (CapEx) into a scalable, low-risk Operational Expenditure (OpEx).

This shift allows organizations to adopt a just-in-time infrastructure strategy. Processing capacity is provisioned only when a revenue-generating project is secured. If no projects are secured, the cost to the firm is zero, a level of financial agility that is impossible with on-premise infrastructure. This economic transformation enables organizational behaviors that were previously impractical:

- Risk-Free Service Expansion: Firms can experiment with offering photogrammetry without facing capital risk.
- Increased Frequency: Inspections can occur quarterly rather than biannually, as resources are no longer scarce.

To visualize this strategic advantage, Table 4 contrasts the operational characteristics of the proposed architecture against standard industry solutions.

Table 4. Strategic comparison of deployment models.

Feature	Traditional Workstation Workflow	Proposed Cloud-Native SaaS Hub
Financial Model	CapEx (High Risk): Upfront hardware purchase required regardless of revenue.	OpEx (Zero Risk): Pay-per-use only when revenue is generated.
Barrier to Entry	High: Requires specialized GPU hardware and software licenses.	None: Accessible via standard web browser on any laptop.
Scalability	Vertical (Limited): Bottlenecked by local GPU/RAM. One project at a time.	Horizontal (Infinite): Auto-scales to handle multiple concurrent projects.
Collaboration	Siloed: Data trapped on local disk; requires manual file transfer.	Unified: Shared URL for real-time multi-user access.
Data Security	Local: Vulnerable to disk failure or local theft.	Cloud: Redundant storage (S3) and encrypted access.

To ground the economic argument in concrete terms, consider the cost structure of a representative small inspection project: a single building surveyed with a consumer-grade drone, producing approximately 150 images, processed to a defect-annotated 3D model. Under the traditional workstation model, the fixed costs include hardware amortization (a workstation with an NVIDIA RTX-class GPU depreciates at approximately USD 150–200 per month assuming a 3-year lifespan and USD 5000–7000 purchase price), software licensing (commercial photogrammetry licenses range from USD 350 to USD 3500 per year), and operator time for local pipeline management.

Under the proposed serverless model, the compute cost for the same project based on the measured processing times reported in Table 3 and Google Cloud Run pricing at the time of writing (approximately USD 0.00002400 per vCPU-second and USD 0.00000250 per GB-second) amounts to approximately USD 0.80–1.20 per project at the container specifications used in this evaluation. For an SME processing 10 projects per month, the annual serverless compute cost is therefore on the order of USD 100–150, compared to hardware and software fixed costs exceeding USD 5000 per year under the traditional model.

While these figures are illustrative rather than exhaustive, network egress, storage, and orchestration costs add overhead, and desktop workflows avoid per-project compute fees; the order-of-magnitude cost structure difference is robust to reasonable variations in these parameters. Critically, the serverless model incurs zero cost during months with no billable projects, a property that the fixed-cost workstation model structurally cannot replicate and that directly addresses the project-pipeline uncertainty faced by SMEs in emerging markets.

While compute costs are dynamically bound to active project execution, the accumulation of high-resolution drone imagery presents a latent long-term Operational Expenditure

(OpEx) risk if not actively managed. A representative inspection project (e.g., 150 images at 10 MB each) generates approximately 1.5 GB of raw optical data. Under standard high-availability cloud storage tiers (e.g., ~USD 0.020 per GB/month), permanently hosting 1000 historical projects would incur a recurring baseline cost of USD 30 per month. To mitigate this and preserve the platform's economic agility, the architecture assumes a tiered data lifecycle policy. Upon project closure, the lightweight 3D mesh, defect metadata, and spatial anchors which are required for ongoing PostGIS spatial queries remain in standard, fast-access storage. Conversely, the heavy raw image datasets are automatically transitioned to "cold" archive storage (e.g., Google Cloud Archive at ~USD 0.0012 per GB/month). Under this lifecycle model, the long-term archival storage cost for the same 1000 historical projects drops to under USD 2.00 per month, demonstrating that storage OpEx can be aggressively contained without sacrificing historical asset intelligence.

4.3. Integration with Existing BIM Ecosystems

A frequent question is how the spatial intelligence hub relates to existing Building Information Modeling (BIM) platforms. We emphasize that the system is complementary rather than competitive. BIM excels at design-phase information management where building geometry is authored through parametric modeling and enriched with semantic specifications. This design-centric information flows forward through construction for coordination.

However, BIM platforms assume information originates during design and struggle with post-construction scenarios where as-built reality must be captured through sensor data. While scan-to-BIM workflows exist for converting point clouds to parametric models, these require substantial manual effort (20–40 h of modeling per building) and often discard high-frequency geometric detail present in photogrammetric reconstructions. The spatial intelligence hub addresses this gap by accepting that post-construction reality often diverges from design intent and therefore requires direct geometric capture rather than parametric reconstruction.

The photogrammetric mesh serves as a bottom-up representation of as-built conditions onto which inspection data is anchored. This as-built Digital Twin complements top-down BIM models where they exist and provides comparable spatial organization benefits for buildings lacking BIM documentation, establishing the framework illustrated in Figure 11. Interoperability is achieved through standard data exchange formats. Defect records can be exported as IFC (Industry Foundation Classes) spatial elements for import into BIM authoring tools. 3D tiles can be overlaid with BIM models in web-based viewers for as-designed vs. as-built comparison.

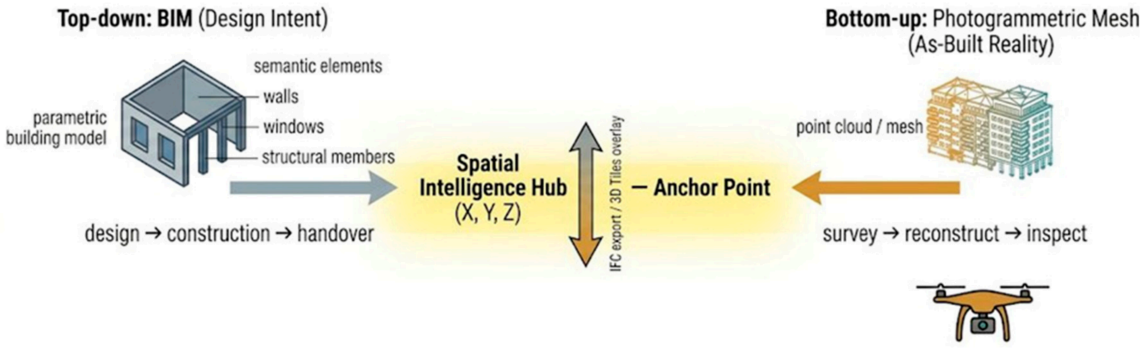

Figure 11. Complementary relationship between top-down BIM and bottom-up photogrammetric reconstruction. BIM encodes design intent and flows forward through construction; photogrammetric mesh captures as-built reality from drone survey. The spatial intelligence hub occupies the intersection, providing a georeferenced anchor layer (X, Y, Z) that bridges both representations through standard exchange formats (IFC, OGC 3D Tiles).

Camera poses and defect locations can be exported as COBie (Construction Operations Building information exchange) spreadsheets for integration with CMMS (Computerized Maintenance Management Systems). This API-first architecture prevents vendor lock-in and enables gradual adoption within existing tooling ecosystems.

4.4. Limitations and Boundary Conditions

Several technical limitations constrain the system's applicability:

Photogrammetric Reconstruction Requirements. Structure-from-Motion algorithms require sufficient visual texture variation for feature matching. Surfaces with uniform appearance (e.g., blank concrete panels, monotone metal cladding) or highly repetitive patterns (e.g., grid facades, brick coursing) can challenge reconstruction. Highly reflective or transparent materials (glass curtain walls, polished stone) violate brightness constancy assumptions underlying photometric stereo. These failure modes can be mitigated through targeted data collection strategies (e.g., adding artificial texture markers, capturing multiple exposures) but represent inherent constraints of vision-based reconstruction.

Beyond complete reconstruction failure, partial data quality degradation presents a more common operational risk. Low-contrast or motion-blurred images reduce SIFT key point repeatability, increasing the proportion of images that fail geometric verification and are excluded from the reconstruction graph. The system's event-driven retry logic handles this gracefully at the feature-matching stage by relaxing match thresholds, but does not currently detect or report which images were excluded, leaving the inspector unaware of coverage gaps. A coverage gap visualization overlaying the reconstructed mesh with a per-region confidence map derived from the number of contributing source images has been identified as a necessary addition before operational deployment.

Network Connectivity Dependency. The cloud-native architecture assumes reliable internet connectivity for uploading imagery and accessing reconstructions. In remote locations or during field work in areas with poor cellular coverage, this dependency creates workflow friction. Hybrid architectures incorporating local caching or offline-capable mobile applications could address this limitation but would increase implementation complexity. For the target use case of urban infrastructure inspection in emerging markets with expanding 4G/5G coverage, this limitation is increasingly manageable.

Scalability Ceilings. While the serverless architecture scales effectively for moderate workloads (dozens of concurrent projects), theoretical limits exist. Database query performance degrades for extremely large portfolios (hundreds of thousands of defect records) without careful index tuning and query optimization. Long-term archival storage costs for high-resolution imagery could become significant for organizations managing hundreds of buildings inspected repeatedly over decades. Intelligent data lifecycle management strategies (automated down-sampling of historical imagery, migration to archival storage tiers, aggregation of temporal data) would be necessary for enterprise-scale deployment.

Absence of Automated Defect Detection. The current implementation requires human inspectors to manually create defect annotations after visually examining reconstructions and source imagery. While this manual workflow reflects standard practice in current inspection operations and was a deliberate design choice, robust spatial anchoring infrastructure must precede AI integration, as a detection model is only as useful as the information substrate to which it writes; it represents the most consequential bottleneck for scaling inspection throughput. The spatially anchored defect database generated by human experts through this platform constitutes precisely the kind of georeferenced, evidence-linked ground truth dataset that supervised defect detection models require but rarely have access to in structured form. Existing defect detection research typically operates on unstructured image collections without spatial context, limiting the ability to train

models that reason about defect location, progression, or spatial relationship to structural elements [5,32]. The present system's output defects annotated with 3D coordinates, severity classifications, linked source images, and supporting documentation are natively structured for training spatially aware detection models. This creates a self-reinforcing improvement cycle: human-in-the-loop annotation through the platform generates training data; training data enables automated pre-annotation; automated pre-annotation reduces inspector workload while maintaining human verification. The integration of automated detection is therefore identified not merely as a feature addition but as an architectural inevitability that the spatial information infrastructure of the present system is designed to support.

Parameter Sensitivity and Hyperparameter Optimization. The current evaluation establishes the end-to-end functional feasibility of the architecture using baseline algorithmic parameters. However, we have not yet conducted a formal sensitivity analysis on the hyperparameter configurations governing the spatial anchoring and reconstruction engines. For instance, the α parameter within the frustum-based image retrieval scoring function (Algorithm 1), which balances the weight of surface normal incidence against Euclidean distance, was empirically fixed rather than systematically optimized. Similarly, the PatchMatch Multi-View Stereo configuration (e.g., the number of depth hypotheses and propagation iterations) was held constant across both datasets to isolate the impact of acquisition geometry on compute time. Because modifying these variables directly affects both the accuracy of spatial queries and cloud resource consumption, a comprehensive ablation study and hyperparameter sensitivity analysis represent a critical next step to mathematically optimize the platform's operational efficiency.

4.5. Threats to Validity

The measured end-to-end runtime depends on cloud execution conditions such as cold starts and multi-tenant resource contention. Although the orchestration design mitigates timeouts, execution time may vary across deployments.

Construct validity: Image retrieval functional correctness is evaluated by manual verification across 30 sampled anchor points, which may introduce subjectivity in borderline cases such as corners and repetitive textures. The evaluation sample is sufficient to demonstrate functional correctness of the mechanism but insufficient to establish statistical performance bounds for operational deployment. A top-k evaluation across a larger and more diverse sample is identified as necessary future work to support claims of retrieval reliability across varying building typologies.

External validity: To properly contextualize the empirical evaluation, the system's performance bounds must be evaluated within the scope of system architecture design rather than a population benchmark study. Accordingly, we satisfied the validation criteria by clearly defining our selection method as a controlled, purposeful contrast rather than a statistical population sample. The two-dataset configuration was specifically engineered to capture opposite ends of the geometric spectrum (nadir-dominated grid vs. circular oblique convergent networks) to evaluate our dynamic serverless resource-provisioning hypotheses. However, this deliberate design choice inherently limits broad generalizability claims, as the system has not yet been stress-tested on asset classes characterized by challenging photogrammetric boundary conditions, such as glass curtain walls, open-frame steel structures or underground infrastructure. Validating the architecture across at least two additional complex building typologies, specifically a masonry heritage structure and a steel-frame industrial facility, has been established as an immediate follow-on roadmap item to systematically map external operational bounds.

4.6. Generalizability to Other Construction Monitoring Applications

The architectural patterns demonstrated in this work extend beyond building facade inspection to other construction monitoring scenarios requiring spatial organization of heterogeneous data:

Progress-Tracking During Construction. Periodic drone surveys of active construction sites could be anchored to a common reference geometry, enabling automated change detection and schedule deviation analysis. The same frustum culling mechanism could retrieve source images documenting specific construction activities for verification against approved shop drawings.

Cultural Heritage Documentation. Historic preservation workflows require detailed condition monitoring of irreplaceable structures. The spatial intelligence hub could organize multi-modal sensor data (photogrammetry, thermal imagery, ground-penetrating radar) around a unified 3D baseline, supporting conservation planning and disaster recovery preparation.

Infrastructure Resilience Monitoring. Critical infrastructure (bridges, dams, tunnels) subjected to periodic inspections could accumulate temporal layers of spatially anchored condition data. Change detection across inspection campaigns would reveal degradation trends and inform predictive maintenance models, transitioning from reactive repairs to proactive intervention strategies.

Post-Disaster Damage Assessment. Following earthquakes, floods, or other disasters, rapid deployment of drone surveys could populate spatial databases with damage annotations. The portfolio-scale query capabilities would enable prioritizing emergency response based on structural criticality and damage severity, supporting evidence-based resource allocation decisions.

The common thread across these applications is the same problem this paper addresses: sensor data that generates observations about physical objects, where the observations need to be organized by their spatial relationship to those objects rather than by the file system structure in which they were stored.

5. Conclusions

This paper has presented the design, implementation, and validation of a cloud-native spatial intelligence hub that addresses data fragmentation in infrastructure inspection through serverless 3D reconstruction and context-aware information management. The work makes five principal contributions to construction informatics scholarship:

First, we demonstrate that computationally intensive photogrammetry pipelines can be successfully decomposed into event-driven serverless workflows, eliminating the need for dedicated hardware infrastructure. The Inngest-orchestrated architecture handles the inherent timeout challenges of long-running SfM/MVS processing while providing robust failure recovery and cost efficiency. Processing cost per typical inspection project represents an order-of-magnitude reduction compared to workstation depreciation and operational costs.

Second, we formalize and implement the spatial anchor mechanism that creates bidirectional linkage between database records and three-dimensional geometric locations. The combination of raycasting for coordinate extraction, frustum culling for image retrieval, and PostGIS spatial indexing for portfolio-wide queries transforms passive 3D visualizations into active information-management substrates. This architectural pattern is generalizable beyond facade inspection to other construction monitoring applications requiring spatial organization of heterogeneous data.

Third, we demonstrate how cloud-native serverless architectures fundamentally transform the economic model of infrastructure inspection from capital-intensive to consumption-based deployment. By eliminating requirements for specialized hardware

and perpetual software licenses, the architecture lowers barriers to entry for SMEs in emerging markets. This shift from upfront Capital Expenditure to pay-per-use Operational Expenditure represents a technological leapfrogging opportunity, enabling organizations to adopt advanced photogrammetry capabilities aligned with actual project revenue rather than speculative infrastructure investment.

Fourth, we release the complete implementation as open-source software, providing a reproducible reference architecture that other researchers and practitioners can validate, extend, and adapt to domain-specific requirements. This contribution to open construction informatics enables collaborative development and prevents vendor lock-in that has historically fragmented the AEC software ecosystem.

Fifth, we provide an empirical characterization of acquisition geometry as the major driver of MVS complexity in a serverless cloud context, demonstrating that a dataset with fewer images and coarser Ground Sampling Distance can require significantly more processing time than a larger, higher-resolution dataset when oblique convergent geometry produces higher baseline-to-depth ratios. While this finding is consistent with established photogrammetric theory, its implications for cloud-native reconstruction pipelines have received little attention; thus, it provides empirical grounding for the checkpoint-based orchestration design and establishes acquisition geometry metadata as a necessary input to cloud resource allocation decisions.

The spatial intelligence hub enables several organizational capabilities for resource-constrained engineering firms:

Portfolio-Scale Asset Intelligence. By standardizing spatial data schemas across all inspected buildings, facility managers can execute analytical queries spanning entire portfolios. Questions like "which building cohorts exhibit similar crack propagation patterns suggesting systemic construction defects" or "where should we prioritize structural strengthening given limited capital budgets" become answerable through database queries rather than requiring manual correlation across disconnected reports.

Evidence-Based Maintenance Planning. The persistent spatial anchoring of all inspection findings creates an evolving historical record of asset condition. Temporal change detection across repeated inspection campaigns reveals deterioration rates and validates maintenance intervention effectiveness. This evidence base supports transitioning from reactive repairs (responding to visible failures) to predictive maintenance (intervening before failures occur based on degradation trends).

Collaborative Multi-Stakeholder Workflows. Real-time synchronization of defect annotations and document attachments enables distributed teams to collaborate effectively without geographic collocation. Field inspectors, structural engineers, facility managers, and external consultants can jointly review inspection findings, annotate observations, and make decisions within a shared spatial interface. This reduces coordination overhead and eliminates information loss from handoffs between stakeholders.

Reduced Barrier to Digital Adoption. The zero-installation browser-based interface and pay-per-project pricing model make advanced photogrammetry capabilities accessible to organizations that would never justify purchasing desktop software licenses or high-end workstations. This democratization is particularly impactful for SMEs in emerging markets, potentially catalyzing broader digital transformation of regional construction sectors.

Several research directions emerge from this work that would extend capabilities while maintaining the core architectural principles:

- Automated defect detection: The spatially anchored defect database generated by human inspectors provides a natural ground truth corpus for training computer vision models. Future work should explore integrating automated pathology detection into the platform, using both source imagery and 3D geometric context as inputs. Robust

spatial information infrastructure must precede AI integration; the present system provides that foundation.

- Temporal change detection: Anchoring repeated inspection campaigns to a common reference geometry would enable automated monitoring of structural deterioration over time. This would support a transition from reactive maintenance to predictive intervention, with degradation trends derived from accumulated spatial observations rather than periodic manual comparison of disconnected reports.
- Advanced visualization with 3D Gaussian Splatting: While SfM meshes provide the geometric reliability required for spatial anchoring, 3D Gaussian Splatting offers superior photorealistic rendering that could improve the inspection review experience. A hybrid architecture using the SfM mesh as the geometric substrate for raycasting while rendering through a Gaussian Splatting layer would preserve spatial accuracy while significantly enhancing visual quality.
- Mobile and offline capabilities: The current architecture assumes reliable internet connectivity, which may be limited at remote inspection sites. Offline-capable mobile applications that cache project data locally and synchronize when connectivity is restored would extend the platform's reach to field environments where the cloud-native model currently creates friction.
- Standardization and interoperability: The spatial anchor concept could inform development of open data exchange standards for inspection information, in collaboration with bodies such as ISO, buildingSMART, and OGC. Standardized schemas for spatially anchored defect records and documented API specifications would enable interoperability across platforms and prevent the vendor lock-in that has historically fragmented construction software ecosystems.

While the current architecture establishes a foundational spatial intelligence hub, future work should focus on integrating three advanced capabilities into the serverless pipeline. First, automated defect detection should be integrated by deploying asynchronous AI inference containers triggered immediately following the final densification stage. The primary challenge here lies in managing the cold-start latency of loading heavy vision-model weights within stateless environments. Second, enabling temporal change monitoring will require the pipeline to automatically co-register independent, multi-temporal drone flights. This introduces the challenge of establishing stable 3D invariant coordinate frames to accurately calculate mesh deformation over time. Finally, the integration of 3D Gaussian Splatting (3DGS) has been proposed to enhance photorealistic visualization. Because native 3DGS lacks the explicit polygonal geometry required for the frustum raycasting described in this study, the proposed integration scheme involves a hybrid rendering architecture: utilizing the explicit SfM mesh as an invisible structural layer for coordinate calculations, while overlaying a real-time rasterized 3DGS layer for user visualization.

It is important to characterize precisely what this work demonstrates and what it does not. The experimental evaluation validates end-to-end feasibility of the proposed architecture on two real UAV inspection datasets representing geometrically contrasting acquisition conditions, and confirms sub-second spatial query performance and greater-than-90% image retrieval accuracy at the building scale. These results establish that the system works correctly under the tested conditions. They do not establish statistical performance bounds across diverse building typologies, acquisition equipment, or geographic conditions; claims of that nature would require a benchmark study across a substantially larger and more varied dataset, which is identified as the primary direction for future empirical work. The contribution of this paper is therefore best characterized as a reproducible reference architecture with demonstrated feasibility, empirically grounded design rationale, and a clearly defined evaluation roadmap. This framing is consistent with the role of

system architecture papers in construction informatics, which establish the design space and demonstrate viability before large-scale benchmark studies become tractable [3,33].

The fragmentation of inspection data represents a systemic risk to infrastructure safety and operational efficiency. While computer vision research pursues increasingly sophisticated detection algorithms and rendering techniques advance toward photorealistic virtual reality, practitioners struggle with more fundamental challenges around information organization, multi-stakeholder collaboration, and portfolio-scale asset intelligence.

This work demonstrates that thoughtful system architecture addressing these workflow challenges can deliver substantial practical value independent of algorithmic sophistication. By making 3D Digital Twins accessible through serverless economics and usable through spatial information management, the spatial intelligence hub charts a path toward portfolio-scale infrastructure intelligence in emerging markets.

As construction organizations in regions like Morocco and the broader Global South continue, digital transformation journeys, the choice between replicating developed-market software accumulation patterns and leapfrogging to integrated cloud-native platforms will have lasting implications. The spatial intelligence hub represents one instantiation of the leapfrogging strategy, and we hope this work encourages broader reflection on how construction informatics can evolve to serve resource-constrained contexts rather than merely transplanting approaches optimized for capital-rich markets. The open-source release accompanying this paper invites the research community to validate, critique, and extend this work.

Construction informatics advances not through isolated algorithmic breakthroughs but through collective development of robust architectural patterns, standard protocols, and open implementations that enable ecosystem growth. We welcome collaboration toward that shared goal.

Supplementary Materials: Source code, deployment documentation, and demonstration. datasets are available at https://github.com/youssef-gis/cloud-native-construction-informatics (accessed on 18 March 2026).

Author Contributions: Conceptualization, Y.A., F.A.-T. and H.B.; Methodology, Y.A.; Software, Y.A.; Validation, Y.A.; Formal Analysis, Y.A.; Investigation, Y.A.; Data Curation, Y.A.; Writing—Original Draft Preparation, Y.A.; Writing—Review and Editing, Y.A., F.A.-T. and H.B.; Visualization, Y.A.; Supervision, H.B. and F.A.-T.; Funding Acquisition, F.A.-T. All authors have read and agreed to the published version of the manuscript.

Funding: This research was partially funded by the European Commission through the DIGIS3 (Smart, Sustainable and Cohesive Digitalization) Digital Innovation Hub project, grant number 101083898-DIGIS3 (2023–2026).

Data Availability Statement: The original contributions presented in this study are included in the article. Source code and demonstration datasets are openly available at: https://github.com/youssef-gis/cloud-native-construction-informatics (accessed on 18 March 2026).

Acknowledgments: The authors acknowledge the open-source communities behind OpenSfM (v0.5, Mapillary AB, Malmö, Sweden, accessed 12 January 2026), CesiumJS, Next.js, PostGIS, Inngest, and Supabase.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AEC	Architecture, Engineering, and Construction
BIM	Building Information Modeling

CapEx	Capital Expenditure
CDE	Common Data Environment
GSD	Ground Sampling Distance
MVS	Multi-View Stereo
O&M	Operation and Maintenance
OpEx	Operational Expenditure
RLS	Row-Level Security
SfM	Structure-from-Motion
SME	Small and Medium Enterprise

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